

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

Total Marks:25

Comparative Analysis

Code of Conduct:

I P. Greeshma (192424418) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATING GRADES

Criteria	Weightage	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Understanding of Concepts	15%	Provides a comprehensive understanding of all concepts being compared	Provides a good understanding of all concepts being compared	Provides a basic understanding of all concepts being compared	Provides a limited understanding of all concepts being compared
Comparison Depth	20%	Provides detailed, insightful and meaningful comparisons across multiple aspects	Provides insightful and meaningful comparisons across multiple aspects	Provides basic comparisons across multiple aspects	Provides limited comparisons across multiple aspects
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Selects relevant comparison parameters	Selects basic comparison parameters	Selects limited comparison parameters
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	Limited analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	Limited conclusion

Topology

Power

Cabling

Required
basic key

could

works with

one cable.

every device

central switch

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RUBRICS FOR CALCULATING

Criteria	Weightage	Weightage
Understanding of Concepts	15%	Provides identified insightful and meaningful comparisons across multiple aspects
Comparison Depth	20%	Selects highly relevant and appropriate comparison parameters
Criteria Selection	10%	Strong analysis with logical reasoning and clear justification
Analytical Skills	15%	Well-organized, clear, and logically structured
Organization & Structure	10%	Uses strong, relevant examples or data to support comparison
Use of Examples / Evidence	10%	Very clear, concise, and easy to understand
Clarity of Presentation	10%	Insightful conclusion with strong justification
Conclusion & Insights	10%	

ogy :-

Topology	Bus Topology
	Lower
	less cabling
	Not required in classic bus
	Lower
	Difficult
	Decreases with Backbone cable

because every device is connected to the central switch

Fault Tolerance
 because multiple devices can potentially be replaced without affecting the network tolerance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Star vs Bus Topology :-

Parameter	Star Topology	Bus Topology
Cost	Higher	Low
Cabling	More cabling required	Less cabling
Central Device	Required	Not required
Reliability	Higher	Lower
Fault isolation	Easy	Difficult
Performance	Better	Decreases with Backbone cable
Main failure	central switch	

Analysis :-

Star topology is more expensive because every device requires a separate connection to the central switch.

2. Mesh vs Star Topology - Fault Tolerance

Mesh topology has higher fault tolerance because multiple paths are available for communication. If one connection fails, another path can potentially be used.

Star topology provides moderate fault tolerance.

1. Star vs Bus Topology :-

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Analysis :-

Star topology is more expensive because every device requires a separate connection to the central switch.

2. Mesh vs Star Topology - Fault Tolerance

Mesh topology has higher fault tolerance because multiple paths are available for communication.

If one connection fails another path can potentially be used.

Star topology provides moderate fault tolerance.

Failure of one host or its cable does not normally affect other hosts. However, failure of the central switch can affect the entire network.

Therefore :-

Mesh :- Very high fault tolerance

star :- Moderate fault tolerance

3. Bus Topology Vs Mesh Topology

Bus topology is relatively simple to install because devices share a common backbone cable.

Mesh topology is much more complex because a full mesh requires direct connections between every pair of nodes.

for n nodes a full mesh requires

$$\frac{n(n-1)}{2} \text{ links}$$

For Example :-

4 nodes $\rightarrow$ 6 links

5 nodes $\rightarrow$ 10 links

10 nodes $\rightarrow$ 45 links.

Therefore, as the number of devices increases, mesh topology becomes increasingly complex and expensive.

Bus topology is easier and cheap to install whereas mesh topology provides better redundancy but requires significantly more cables, ports and configuration.

Hybrid Topology Vs Star topology - Scalability

Hybrid topology is generally more scalable because it combines multiple topology types.

For Example:-

Mesh / Ring → campus backbone

Star → individual buildings

This allows different sections of the network to be expanded according to their requirements.

Star topology is also scalable, but its expansion depends on the available ports and capacity of the central switch. When all ports are occupied additional networking equipment may be required.

Therefore hybrid topology provides greater flexibility and scalability, especially for large organizations such as universities and enterprises.

Maintenance Comparison

Star:-

Star topology is easiest to maintain because each device has an individual connection to the central

Switch A faulty device

Mesh:-

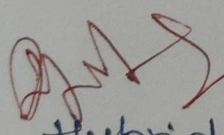
Mesh topology is highly reliable but difficult to maintain because it has many connections.

Bus:-

Bus topology is harder to troubleshoot because a problem in the backbone cable can affect many devices

Hybrid:-

Hybrid topology has moderate maintenance complexity.

Topology	Maintenance level	Reason
Star	Easy	Individual connections make fault easy to isolate
Mesh	Difficult	Large no. of connections
Bus	Difficult	Backbone faults are difficult to locate
 Hybrid	Moderate	Multiple topology structures require skilled management